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Arbitrage, Replication, and the Cost of Carry in Pricing Derivatives

REPLICATION

Learning Outcome

- explain how the concepts of arbitrage and replication are used in pricing derivatives

Replication is a strategy in which a derivative's cash flow stream may be recreated using a combination of long or short positions in an underlying asset and borrowing or lending cash. In contrast to the earlier arbitrage examples, replication is typically used to mirror or offset a derivative position when the law of one price holds and no riskless arbitrage profit opportunities exist. For example, Exhibit 4 compares a long forward commitment to the alternative of borrowing funds at the risk-free rate, r, to buy the underlying asset at today's spot price, S_0.

Exhibit 4:

Forward Commitment Replication

Borrow S_0 and purchase at spot (S_0)	**Repay loan ($S_0(1 + r)^T$) and sell at**
Borrow USD1,783.28 (S_0) today at $t = 0$	Repay USD1,792.13 ($S_0(1 + r)^T$) a
Buy at S_0 = USD1,783.28	See at spot (S_T)
CF_0 = (USD1,783.28 – USD1,783.28) = 0	**CF_T = (S_T – $1,792.13)**
Enter long forward contact ($F_0(T)$)	**Settle long forward contact (S_T**
Agree to buy at USD1,792.13 ($F_0(T)$) at $t = T$	Settle contract ($S_T - F_0(T)$) at
CF_0 = 0	**CF_T = (S_T – $1,792.13)**

0 — Time — T

▶ Expand / collapse Extended Description

EXAMPLE 2

In Example 1, we reintroduced Procam Investments, which borrowed and purchased gold in the spot market and simultaneously sold gold in the forward market to earn a riskless profit. Here we change the assumption about today's spot gold price (S_0). Specifically, the spot gold price has risen to USD1,783.28 (= USD1,792.13$(1.02)^{-0.25}$) to eliminate the earlier arbitrage opportunity when the spot price was USD1,770. Assume again that the risk-free interest rate (r) is 2% and gold can be stored at no cost.

1. Long forward commitment (agree to buy at $F_0(T)$ at $t = T$)
   - Procam enters a forward commitment to buy 100 ounces of gold in three months at a forward price, $F_0(T)$, of USD1,792.13 per ounce.
   - If S_T = USD1,900 per ounce,
     - Profit = USD10,787 = 100 × (USD1,900 – USD1,792.13).
   - If S_T = USD1,700 per ounce,
     - Profit = –USD9,213 = 100 × (USD1,700 –USD1,792.13).
2. Borrow and purchase (borrow S_0 and buy asset at S_0 at $t = 0$):

- Procam repays the loan principal and interest: USD179,213 = USD178,328$(1.02)^{0.25}$.
- If S_T = USD1,900,
  - Profit = USD10,787 = 100 × (USD1,900 – USD1,792.13).
- If S_T = USD1,700,
  - Profit = –USD9,213 = 100 × (USD1,700 – USD1,792.13).

Note that the same profits are observed under Scenarios 1 and 2, which are equal to $S_T - F_0(T)$. If we set the forward price, $F_0(T)$, equal to the future value of the spot rate using the risk-free rate ($F_0(T) = S_0(1 + r)^T$), the no-arbitrage condition demonstrates that Procam generates the same cash flow at time T regardless of the direction of gold prices whether the company

1. enters into a long forward commitment settled at time T or
2. borrows at the risk-free rate, buys the underlying asset, and holds it until time T.

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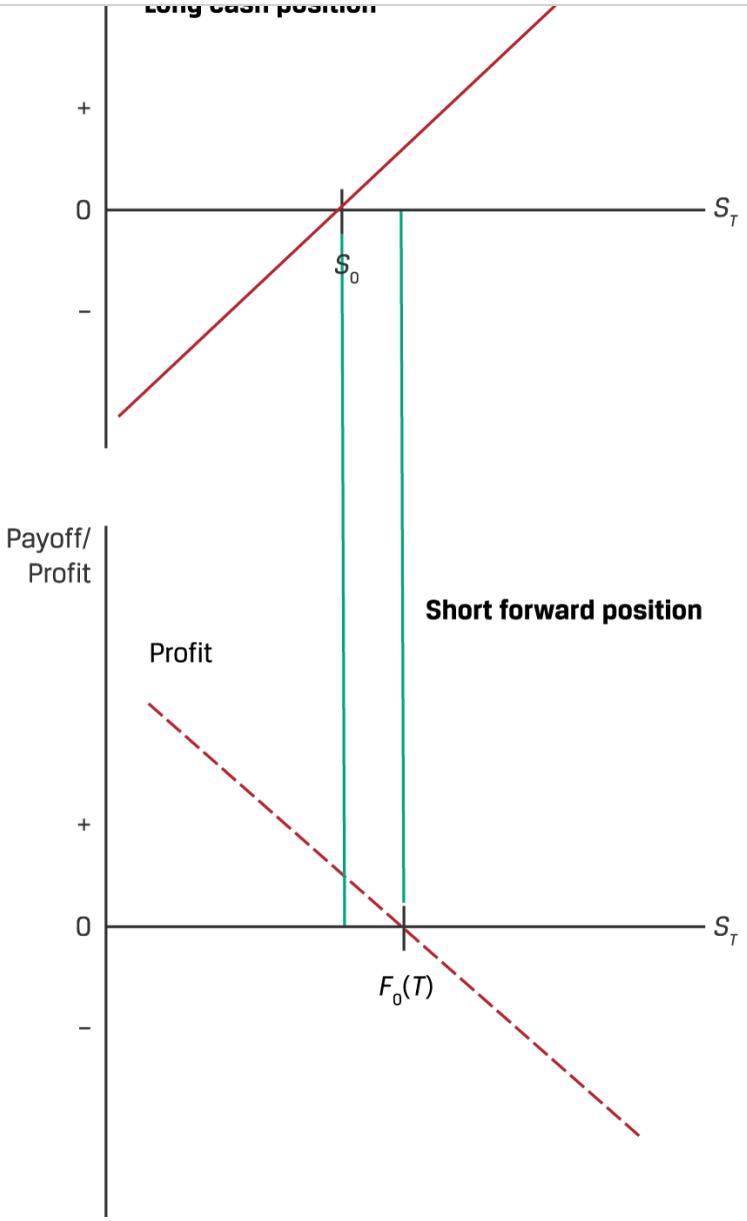
A separate but related form of replication pairing a long asset with a short forward is shown in Exhibit 5. The top half shows the profit/loss profile of the long asset, and the bottom half shows that of the short forward position. The long asset position produces a profit (loss) when $S_T > S_0$ ($S_T < S_0$). The short forward position produces a profit (loss) when $S_T < F_0$ ($S_T > F_0$).

Exhibit 5:

Payoffs for Long Asset and Short Forward Positions



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In contrast to Example 2, these positions appear to offset one another. The following example evaluates the return generated by these combined positions.

EXAMPLE 3

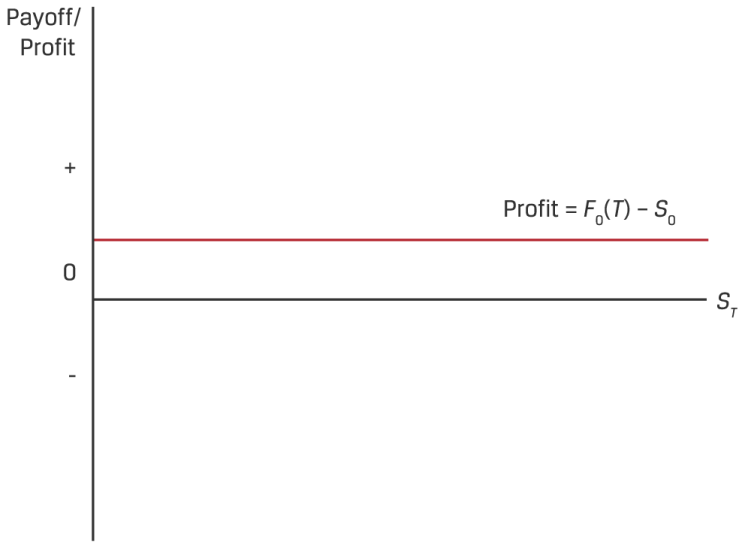
Procam Investments buys 100 ounces of gold at today's spot price (S_0) of USD1,783.28 and simultaneously enters a forward commitment to sell gold at the forward price, $F_0(T)$, of USD1,792.13. Again, we assume that gold can be stored at no cost, and here we solve for Procam's rate of return.

- At $t = 0$, Procam's cash flow is $-S_0$ = –USD178,328.
- At $t = T$, Procam's cash flow is $F_0(T)$ = USD179,213.
- Solve for the rate of return on Procam's strategy, as follows:
  - USD179,213 = USD178,328$(1 + r)^{0.25}$.
  - r = 2.0%, which is equal to the risk-free rate.

This example demonstrates that Procam can hedge its long gold cash position with a short derivative (i.e., selling the forward contract) and earn the risk-free rate of return

Exhibit 6:

Combined Long Asset and Short Forward Profit



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Because the forward price is assumed to be greater than the spot price in this example and r is positive, the risk-free profit is a positive amount and is the same (i.e., risk free) regardless of the price of the underlying (i.e., S_T). As we will see in the next lesson, there may be benefits or costs to owning an underlying asset that cause the risk-free return in Exhibit 6 to differ from $F_0(T) - S_0$ and possibly even be negative.

Question Set: Arbitrage and Replication

1. Determine the correct answers to fill in the blanks: If the law of one price does not hold, a(n) _____ asset trades at the same time at _____ prices in different places.

Hide Solution

Solution:

If the law of one price does not hold, a(n) *identical* asset trades at the same time at *different* prices in different places.

2. An investor observes that the spot price, S_0, of an underlying asset with no additional costs or benefits exceeds its known future price discounted at the risk-free rate, $S_T(1 + r)^{-T}$. Describe and justify an arbitrage strategy that generates a riskless profit for the investor.

Hide Solution

Solution:

Since the spot price of the underlying asset exceeds the known future price discounted at the risk-free rate ($S_0 > F_0(T)(1 + r)^{-T}$), at $t = 0$, the investor:

• Sells the underlying asset short in the spot market at S_0

At time $t = T$, the investor:

- Settles the long forward position and receives $S_T - F_0(T)$
- Offsets the short underlying asset position at S_T, and
- Retains $S_0(1 + r)^T - F_0(T)$ as a riskless profit regardless of the underlying spot price at time T.

3. Formulate a replication strategy for a three-month short forward commitment for 1,000 shares of a non-dividend-paying stock.

Hide Solution

Solution:

The replication strategy for a three-month short forward commitment on a non-dividend-paying stock involves the short sale of 1,000 shares of stock at $t = 0$ and investment of proceeds at the risk-free rate, r. At time $t = T$, the short sale is covered at S_T, and under the no-arbitrage condition of $F_0(T) = S_0(1 + r)^T$, the return is equal to $F_0(T) - S_T$ for both the short forward and the replication strategy.

4. Calculate the arbitrage profit if a spot asset with no additional costs or benefits trades at a spot price of 100, the three-month forward price for the underlying asset is 102, and the risk-free rate is 5%.

Hide Solution

Solution:

The forward price, $F_0(T) = S_0(1 + r)^T$, at which no-arbitrage opportunities would exist is 101.23 (= $100(1.05)^{0.25}$). With an observed forward price of 102, the arbitrage opportunity would be to sell the forward contract and buy the underlying, borrowing at the risk-free rate to fund the purchase. The arbitrage profit is the difference between the observed forward price and the no-arbitrage forward price of 0.77 (= 102 – 101.23).

5. Describe the relationship between the spot and forward price for an underlying asset if the risk-free rate is negative.

Hide Solution

Solution:

The relationship between the spot and forward rate for an asset with no additional costs or benefits of ownership other than the opportunity cost (risk-free rate) is equal to $F_0(T) = S_0(1 + r)^T$. In a case where $r < 0$, $(1 + r)^T < 1$ and therefore the forward price, $F_0(T)$, is below the spot price, S_0, if the risk-free rate is negative.

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